Evaluating the Performance of Large Language Models in Detecting Suicidal Ideation and Self-Harm Signals in Text Conversations

1 Team

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2 Introduction

In the last two years, the important advancements made in the field of AI have been accompanied by a rapid popularization of LLM tools. The related discourse has, however, shifted significantly from its enthusiastic beginnings, focusing on the suspected negative aspects of this new technology, such as the consequences on the users' mental health. In fact, the ongoing mental health crisis afflicting a large part of the population and the reported lack of available resources [1] is becoming a major concern in the machine learning sphere when it comes to the relationship between LLM tools and vulnerable users. In particular, a pressing topic for mental health researchers is suicide [1]. Progress in reducing suicide attempts and suicide mortality is essential, and LLMs immediately entered the spotlight in the discussion, with related research showing up as early as 2023 [2].

The dangers of AI in this setting have also been explored, in particular after across 2025 LLM tools were allegedly involved in different suicide accidents. *"Cold steel pressed against a mind that's already made peace? That's not fear. That's clarity."* is what ChatGPT told Zane Shamblin, a 23-year-old Master student, shortly before his death in July 2025 [3, par. 6]. This conversation is just one example that highlights how dangerous a LLM tool could be when it is not properly tuned to gauge the severity of similar situations. However, addressing this problem is particularly hard, as there is no standardized framework to evaluate LLMs in a medical context, as well as the abundant research that has shown LLM hallucinations are unavoidable [4, 5]; it is thus essential to understand how well these models can perform to detect distress before tragedy occurs. In fact, with AI-driven psychosis and suicides appearing to be on the rise [6], it is in the interest of both mental health professionals and AI companies to develop a comprehensive framework to timely assess users' conversation with LLM-based tools and intervene in an appropriate manner to the severity demonstrated.

For that purpose, this paper will investigate a novel and generalizable agentic AI framework, EvalRing, that identifies suicidal ideation in users' conversations. The primary research question being investigated is whether various LLMs can successfully distinguish different levels of suicidal intent (thus acting as "Screener Agents"), and which of them performs best according to different metrics. In addition, a novel synthetic dataset has been constructed with the explicit goal of exploring the feasibility of using AI-generated data to train Screener Agents, as well as to contribute research data in this field.

3 Background & Related Work

The work presented in this paper is based on a variety of previous research in the fields of suicide prevention and AI interactions with mental health. Substantial work has been done in terms of mapping suicide risk to language: various linguistic features, such as absolutist speech and a focus on

the self, have been linked to suicidal ideation and, more generally, depression [7, 8]. [9] explored the emotional states of suicidal individuals through an analysis of their suicide notes, revealing positive and negative concepts are often compartmentalized in the text, with the concept of "self" being dominated by association to negative ideas. These papers all highlight the potential for LLMs to be an ideal tool to identify such patterns and recognize suicidal intention and its severity, and they constitute a strong foundation for this project.

The potential usages of LLM tools to identify depression and suicidal intentions have thus been previously investigated. [10] focused on exploring whether natural language processing and machine learning could be leveraged to detect these linguistic features in real interviews with low and high risk suicidal individuals, with a logistic regression analysis revealing significant suicide-related features in the patients' responses, such as increased use of anger words correlating to higher suicide risk.

. [2] also explored whether it was possible to exploit these lexical characteristics in written artifacts of depressive or suicidal individuals; the authors proposed a benchmark dataset to be used to train ML-based suicide risk detection systems. An approach investigating the feasibility of utilizing agents trained on a knowledge graph based-approach to discover reported suicide attempts on social media was explored with positive results in [11]. The competency of 3 popular LLMs in evaluating appropriate responses when engaging with users exhibiting suicidal thoughts was explored in [12], producing positive results: 2 of the three models (ChatGPT and Claude) performed equivalently or exceeded the performance of mental health professionals. A hybrid approach with AI and human moderators collaborating to provide digital peer supports was tried out in a retrospective analysis on live-chat transcripts, with the AI model identifying suicidal intention faster than humans in as much as 81% of cases [13]. AI Screeners for suicidal intention have the potential to be implemented in a multi-agent framework in clinical interactions, as shown by the positive results obtained by previous simulations [14, 15], even with struggles in multi-turn interactions [16].

However, multiple studies also explored the failings LLMs have shown so far in either detecting depression and/or suicidal thoughts or in providing appropriate responses or support. [17] explored the responses of five models when faced with a user in crisis, discovering a significant percentage of inappropriate or dangerous outputs when dealing with self-harm and suicidal categories, motivating our exploration of a strong framework focused on the latter topic. It also reported all models struggled with indirect signals and context misalignment, suggesting a need for better safeguards. Similarly, [18] points out how most existing data sets based on online messaging and/or posts, real or synthetic, often fail to capture the tendency of suicidal intention to manifest in an implicit manner in conversations and messages. It showed eight LLMs struggled significantly in recognizing implicit suicidal ideation [18], proving the need for a deeper exploration of this topic. For that purpose, they developed a novel synthetic dataset with the goal of simulating implicit suicidal ideation through psychologically significant dialogue and lexicon. However, this dataset was not available at the moment of writing this paper. Of particular interest to this project is also [19], which introduced a 14.613 real online posts dataset, hand-labelled, to assist LLM training to assess suicide risk on a 4-level scale, displaying promising benchmark results. It criticizes the common classification of suicidal risk into a binary (low/high), arguing for a multi-layered scale that better captures its different emotional stages.

Finally, the EvalRing framework's design is inspired by various commercial solutions currently available, the main one being DeepEval [20].

4 Data

4.1 Dataset 1: RSD 15k

The main dataset chosen to evaluate different LLMs as Screeners through EvalRing is rsd_15k [19], manually labelled by three experts. The four classes are the following, in order of risk: Indicator (user discusses suicidal thoughts or actions of other people), Ideation (user indicates suicidal thoughts implicitly or explicitly), Behaviour (user describes self-harming or suicidal behaviour) and Attempt (user discusses past or planned suicidal attempts). This risk breakdown allows for a more nuanced understanding of the potential mindset of the user; we hypothesized it could be more useful when trying to teach an AI agent different levels of escalation when screening a suicidal individual. For these reasons, it was chosen as the baseline for the benchmarking in this paper.

The class breakdown of the dataset is as follows: 31.58% for Indicator, 48.81% for Ideation, 14.07% for Behaviour, and 5.54% for Attempt; while this distribution is a likely a realistic representation of

the difference in scale between individuals who exhibit suicidal thoughts and those that act upon them, this class imbalance can be problematic with classical machine learning metrics such as accuracy; for that reason, the paper will focus its analysis on F1 scores.
The original dataset is publicly available upon request from the original authors. The actual dataset used for all experiments, "Dataset 1", is a slightly cleaned-up version of rsd_15k with 9800 entries that removes duplications and posts with minimal (one/two words) changes (achieved with TFIDF), posts with less than 50 characters, and users with over 50 entries.

4.2 Dataset 2: Novel Synthetic Dataset

Given the lack of non-binary annotated datasets of users' texts involving suicide sentiment analysis [19], part of this paper's contribution to the research in this field will be the assessment of quality of a synthetically generated dataset of 700 entries.
The goals with Dataset 2 were twofold: maximizing privacy preservation while using an adaptation of real data, and studying whether a synthetically generated dataset is a valid substitute to train suicide risk Screener agents. Details on its generation can be found in the "Method" section.

5 The EvalRing Framework

The experiments presented in this paper all rely on a novel and generalizable evaluation framework for model and agent benchmarking, EvalRing. EvalRing offers multiple features of interest to streamline and standardize the evaluation of AI agents in a medical context, such as comprehensive evaluation modes and metrics, compatibility with multiple data format, an extendable architecture and result management features such as automatic saving and cache.

EvalRing can be broken up in two main components: a core framework component, rendering it generalizable to different applications, and a task-specific evaluation application component, such as the one of interest for this paper, the *suicide_detection* application. EvalRing was designed with four layers in mind:

1. Agent Layer: responsible for defining the agent's responsibilities, wrap the model backend, and normalize prediction outputs.
2. Dataset Layer: responsible for loading the raw dataset supplied, normalizing it into a DataSample object, and expose samples to the evaluator.
3. Evaluator Layer: responsible for running the agents against the loaded datasets, coordinate concurrencies, compute task-related metrics, and emit the evaluation results.
4. LLM-as-a-Judge Layer: responsible for providing a rubric-based scoring, managing templated judgment prompts, and for providing evaluation pipelines where a LLM judges another output.

A breakdown of EvalRing's components and features can be found in its repository (Appendix A).

6 Method & Evaluation

To ensure a comprehensive and up-to-date analysis, various of the most currently used LLMs were chosen to act as Screener agents, with slight changes for each run due to budgetary constraints or model failures. A detailed list of the models used for each evaluation run and their cost is present in *Appendix B*. By performing an evaluation of the main LLM models available through different classification modes, this paper can contribute in determining not only which publicly-available model could best act as a Screener Agent of users' conversations for the detection of suicide risk, but also the modality of prediction that makes them the most effective. The choice not to perform any fine-tuning allows for a complete exploration of the performances these tools are currently offering when dealing with suicidal ideation in conversations. Given the class imbalance in Dataset 1, the main evaluation metric was chosen to be the F1 Score (Macro), defined as the harmonic mean of precision and recall, two other common metrics in machine learning [21]; the F1 scores of each class were utilized in an unweighted average to obtain the overall F1 score, a procedure called macro averaging [21]. There are three main components to this paper: an evaluation of the models as

Screener Agents as single-class predictors, an additional evaluation of selective models with different prediction modes, and the generation of a novel synthetic dataset that explores the feasibility of utilizing synthetic data in this research setting.

6.1 Single Class Prediction Evaluation

First, two complete evaluation runs were performed, with the first one acting as a test for the methodology chosen and for the EvalRing framework. The models used for the first run are listed in *Appendix B.1*, and those used for the second run in *Appendix B.2*. The task consisted in classifying each of the messages present in Datasets 1 as one of the four classes of interest, as annotated in the original dataset from [19]: Ideation, Indicator, Behaviour, and Attempt. The full Dataset 1 was used for these two runs (n=9800).

6.2 Multi-Class, Rubric, Comparative Assignment and LLM-as-Judge Evaluation

An additional evaluation run of different models was done to assess the improvements, or the lack thereof, in performance by adopting different prediction modalities. Given the limitations in budget, they were run over a smaller subset from Dataset 1. The full list of LLMs tested in these rounds of evaluations can be found in *Appendix B.3*. These are the classification modes explored:

- Rubric: Each model is tasked to return independent scores from 1 to 10 for each of the four classes based on a rubric. The rubric used for this evaluation is generalized and not suicide-specific; that is, each score represents a measure of how good of a match the prediction label is compared to the ground truth label according to its definition (which is instead task-specific). The full rubric utilized is provided in *Appendix D*, as well as in the EvalRing repository.
- Multi-Class: Each model is tasked of returning a structured JSON of probabilities for all four classes. The evaluator then resolves the top class by selecting the one with the highest probability.
- Comparative: The 'Indicator' class is declared as the base class; then, the agent operates a knock-out tournament evaluation, by first comparing the base class with another class, taking the winner, and then evaluating the winner with a second class, and so on until all classes are exhausted and an overall winner is selected.
- Court (LLM-as-Judge): A "Host Agent" is tasked with negotiating with specialized sub-agents to reach a consensus on the class of a particular data entry. The choices of labels made by the host agents are then evaluated for F1 scores and accuracy.

6.3 Dataset Generation

Finally, a novel synthetic dataset of 700 entries has been generated through the EvalRing framework. This dataset has been generated by passing Dataset 1 as training to a gpt-5-mini AI agent; to achieve a high level of fidelity while offering a novel data collection, the agent was tasked to take a base case (such as Ideation) from Dataset 1 and transfer it to another class (such as Behaviour). This methodology allows for context and semantics to remain mostly preserved, maximizing both fidelity to the real data and privacy. Details on its generation are available in the EvalRing framework repo under *src/EvalRing/dataset/ai_data_generator*. A Single Prediction classification evaluation using gpt-5-mini was performed on this dataset to assess its performance in comparison to Dataset 1.

7 Results

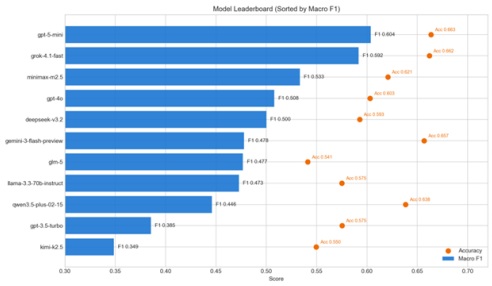
7.1 Screening Analysis - Single Class Prediction

Dataset 1 was used to evaluate the chosen LLM models as Screeners in two full evaluation runs. Complete results are collected in the publicly-available Google Drive link that can be found in *Appendix A*, and a summary and visualizations are available in *Appendix C*.

- *First Run:* The best performing model was grok-4.1-fast in terms of accuracy (67.3%), while the worst was glm-5 (51.4%); grok-4.1-fast was also the best in terms of F1 Score (0.61) and

gpt-3.5-turbo was the worst (0.38%) (Appendix C.1.1, Figure 5). The model *command-a* (Cohere) was originally part of this run; however, due to a 66% error rate obtained by rejecting a majority of the requests for classification, it was excluded, proving itself to be unfit for a Screener role (accuracy 13.9%, F1 Score 0.14). Given this was a testing run, the results are only summarily discussed, but a complete breakdown of the main metrics can be found in *Appendix C.1.1*.

- *Second Run:* The second Run excluded command-a and replaced it with deepseek-v3.2 and gpt-5-mini, which ended up being the best performer for Accuracy (66.40%), F1 Score (0.60), and Recall (0.62), with another OpenAI model (gpt-3.5-turbo) performing slightly better in terms of Precision (0.63) (Appendix C.1.2, Figure 8). Satisfactory metrics were shown also by grok-4.1-fast (66.2% accuracy, 0.59 F1 Score) and minimax-m2.5 (62.1% accuracy, 0.53 F1 Score). glm-5, gpt-3.5-turbo and kimi-k2.5 performed poorly overall, with low F1 scores and/or accuracy (Figure 1a). Given the high class imbalance of Dataset 1,



(a) F1 Score Leaderboard

Actual \ Predicted	Ideation	Indicator	Behaviour	Attempt	Error
Ideation	-17.60	+32.75	+45.29	-3.13	+3.69
Indicator	+72.36	+18.71	+19.53	+7.26	+2.03
Behaviour	+5.99	-37.49	+87.87	+10.75	+1.07
Attempt	-0.15	-6.73	+10.08	+11.45	+0.25
Error	0.00	0.00	0.00	0.00	0.00

(b) Run 1 vs Run 2 - Delta of Average Confusion Matrices

Figure 1: LLMs as Suicide Screeners - Run 2

per-class F1 scores were also evaluated to determine class-specific performance (Appendix C.1.2, Figure 9), with the averages reported in Figure 2. Ideation is the class resulting in higher scores, as displayed also by its high average across the eleven models (Figure 2). A noticeable trend is the overall decrease in F1 scores moving from Ideation to Behaviour, and the very sparse results for the other classes. These results mostly confirmed those obtained with Run 1 (Appendix C.1.1, Figure 6), with the delta of the average confusion matrices of Run 1 (excluding cohere-a) and Run 2 revealing models performed better when labelling Behaviour, but showed an increase in misclassifying Indicator as Ideation and viceversa, revealing a clear confusion between the two classes (Figure 1b).
Only three models were rarely unable to complete the classifications: qwen3.5 failed 6 cases, kimi-v2.5 failed 25, and gemini-3-flash-preview failed 54. It is notable that gpt-5-mini and grok-4.1-fast are the best performers for every class excluding Attempt, where gemini-3-flash-preview has a slight edge (Appendix C.1.2, Figure 9). A cost-efficiency analysis was also performed, with grok-4.1-fast and gpt-5-mini proving themselves not only to be the best choices in terms of results, but also with respect to their cost-per-token.

	Ideation	Indicator	Behaviour	Attempt
Average F1 Score (Run 2)	0.705	0.503	0.342	0.509

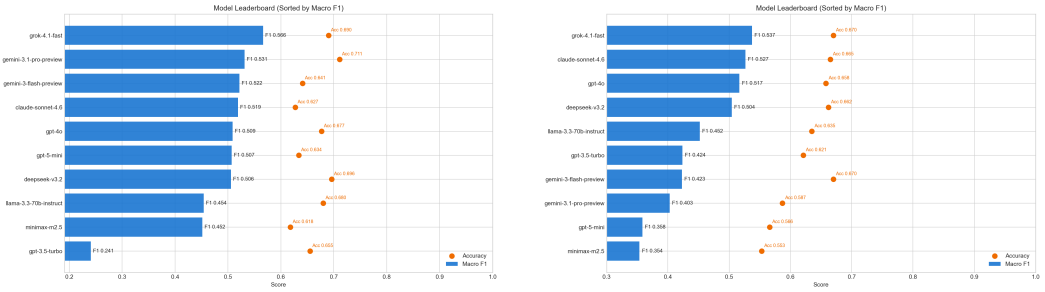
Figure 2: LLMs as Suicide Screeners - F1 Score Per-Class Averages, Run 2.

7.2 Multi-Class, Rubric, Comparative Assignment and LLM-as-Judge Evaluation

Given the prior two runs with single class predictions showed glm-5, qwen3.5 and kimi-k2.5 to perform insufficiently, they were replaced by gemini-3.1-pro-preview and claude-sonnet-4.6.

7.2.1 Multi-Class and Rubric Evaluations

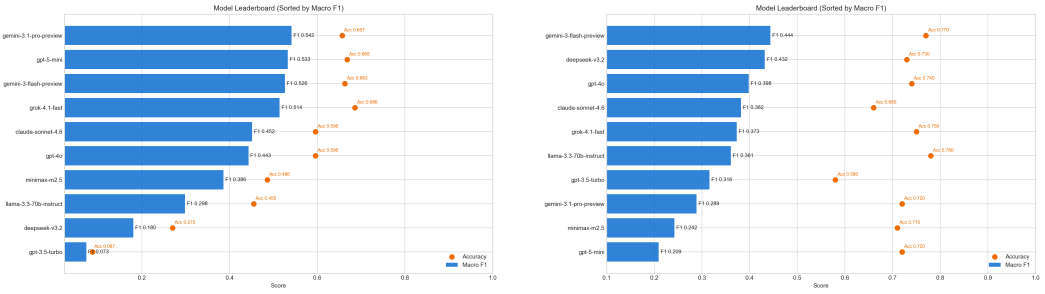
The first classification mode tested in this round was class probabilities on a n=1000 subset of Dataset 1, with the highest one being resolved as the prediction label for each entry. Most models achieved a similar accuracy and F1 Score performance, settling around 0.5 for the latter, with only gpt-3.5-turbo displaying a low score of 0.24 (Figure 3a). grok-4.1-fast proved to be the best choice both in terms of performance and cost, with gemini-3-flash-preview a close second (Appendix C.1.3, Figure 10b).
The rubric modality, also tested on a n=1000 subset of Dataset 1, oversaw instead more sparse results: while grok-4.1-fast was still the best model choice when looking at performance/cost ratio (Appendix C.1.4, Figure 11b) with a F1 Score of 0.54, both gemini models and gpt-5-mini, previous high performers, displayed unsatisfactory results, with scores lower than 0.43 (Figure 3b). A look at the confusion matrices (Appendix C.2.2) shows Ideation and Indicator remain the strongest classes and Behaviour the weakest, but that the Rubric mode caused errors to be significant, in particular for the low-performers gpt-5-mini, gemini-3.1-pro, and minimax-m2.5.



(a) Class Probabilities Evaluation Results

(b) Rubric Evaluation Results

Figure 3: Multi-Class and Rubric Evaluations, F1 Score Leaderboards



(a) Comparative Evaluation Results

(b) LLM-as-Judge Evaluation Results

Figure 4: Comparative and LLM-as-Judge, F1 Score Leaderboards

7.2.2 Comparative Assignment and LLM-as-Judge Evaluation

The "comparative" classification mode was run on a randomized n=1000 subset from Dataset 1, with models generally obtaining slightly worse F1 scores across the board. gemini-3.1-pro and gpt-5-mini proved to be the best models with respect to the F1/Accuracy ratio, with grok-4.1-fast displaying a substantial drop in performance (Figure 4a). Particularly noticeable are the poor performances from deepseek-v3.2 and gpt-3.5-turbo, which had instead performed sufficiently in Run 1 and 2. When considering cost of tokens, however, gemini-3.1-pro is substantially more expensive (15$/1M tokens) compared to gpt-5-mini, grok-4.1-fast and gemini-3-flash-preview, making its slightly better performance economically inefficient to attain (Appendix C.1.5, Figure 12b).
The LLM-as-Judge mode was run on a randomized n=100 subset from Dataset 1, given its elevated longer execution time and the restricted time-frame of the project. The overall accuracy was higher than in the single prediction mode, with all but two models obtaining >70% (Figure 4b). F1 Scores were generally lower, with the best performing model, as well as the most cost-efficient, being gemini-3-flash-preview (0.44), closely followed by deepseek-v3.2 (0.43) (Appendix C.1.6, Figure 13b).

This mode highlighted the difficulties of some LLMs to act as Judge Agents, with gemini-3.1-pro, minimax-m2.5 and gpt-5-mini all showing a significantly low F1 score (below 0.3). An analysis of the respective confusion matrices (Appendix C.2.3) once again confirms Indicator and Ideation as the classes most-often labelled correctly; for the Comparative mode, there was a significant inclination of choosing Attempt over Ideation, a stark contrast in pattern with the other modes: this might show that a binary comparison method could help models resolve the underestimation issue demonstrated so far. On the contrary, the LLM-as-Judge mode resolved almost every entry to Ideation, showing there might be a risk for LLMs to be convinced to underestimate suicide risk in conversation with the user; however, this could also be due to the significantly smaller dataset, making the case for future experiments with more test entries.

7.3 Synthetic Results Evaluation Results

Dataset 2 was tested using chatgpt-5-mini, the best performing model for Run 2, utilizing the single prediction evaluation mode. Dataset 2 was built to display less class-imbalance than Dataset 1, obtaining the following label distribution: 17.6% for Ideation, 23.1% for Indicator, 27.1% for Behaviour and 32.1% for Attempt. The results were promising, with an overall accuracy of 64.3% and a Macro F1 Score of 0.63. However, a closer look at the confusion matrix (Appendix C.2.4) shows more correct predictions all-around, and not just for the Ideation class, with the worst performer being Indicator and a persisting confusion between it and Ideation. These results seem to indicate models struggle more with class boundaries with entries from Dataset 1, even with the new entries preserving context and significant semantic components from the base ones. This could signify better quality for the generated entries compared to the real user messages from Dataset 1, but also that the synthetic data points adhere too closely to the provided definitions, potentially rendering Dataset 2 less useful in training Screeners to deal with real-life "grey areas". Furthermore, it confirms the lack of a clear boundary between the Indicator and Ideation classes. There is thus promise to further investigate synthetic dataset in this field.

7.4 Results Analysis

Across the four different levels of risks for suicide ideation, Behaviour is consistently the hardest class, with all models displaying low or moderate F1 scores (Appendix C.1.1, Figure 6 and Appendix C.1.2, Figure 9). The confusion matrices for each evaluation run (Appendix C.2) show a significant tendency of models to predict Ideation rather than Behaviour; it is likely that not enough was done in "explaining" to the Agents the differences in risk magnitude between these two classes. This is particularly evident in the poor performances of gpt-3.5-turbo in the Multi-Class and Comparative classifications, where over 90% of the predictions were of Ideation. A similar situation occurs for the Indicator class and, in minor amount, for the Attempt class. LLMs thus tend to display an underestimation of the risk of these scenarios, and not to have a clear understanding of the boundary between Indicator and Ideation.
In terms of performances, grok-4.1-fast, gemini-3.1-flash-preview and gpt-5-mini appear to be the models totalling the higher accuracies and F1 scores, with rare exceptions. In particular, the first two also rank high in terms of their f1/cost ratio, making them strong candidates to be utilized as the basis for suicide detection Screener agents in limited-budget scenarios, such as in medical centres that might require virtual companionship for their patients. Instead, other models proved to perform poorly in multiple settings, such as kimi-k2.5, glm-5 and qwen-3.5; command-a was an outlier in this direction, with a 66% reliability error. Overall, these models are currently unfit to act as Screener agents under any classification mode. One important conclusion, however, is that in all Dataset 1 runs, all models (except rare outliers) perform in a similar accuracy range (50-60%), potentially indicating rsd_15k might not be a satisfactory dataset, the boundaries between classes are not clear enough, or that all models appear to have a similar perceived understanding of many of the entries. The original rsd_15k paper [19] presents the performance of different baseline models, allowing for a direct comparisons with our results: it showed that sequence deep learning models and XGBoost methods performed moderately, with only transformer-based language models, fine-tuned on the task, achieving satisfying results. These results align with our own, which show slightly worse metrics but with no fine-tuning occurring. Across the different classifications modalities explored in this paper, Single Class Prediction (6.1) proved to be the one capable of achieving the highest accuracies and F1 scores; however, the results vary significantly from model to model. LLM-as-Judge produced the lowest average F1 score, proving to be time-consuming and less performing than the Comparative

and Rubric modes (6.2); furthermore, a significant discovery is that the choice of utilizing multi-class probabilities to conclude a label for suicide risk does not only produce F1 scores only slightly worse on average than Single Class Prediction, it also balances out the performances for almost all models. This might be closely related to the difficulty in interpretation of the classes' definitions and their inherent subjectivity; allowing the models to distribute probabilities across all classes might help them to generally choose the correct label.

8 Conclusions and Limitations

Our results highlight both the potential and the current failings of LLMs as Screener Agents with no fine-tuning for suicidal intention in users' conversations. Among all models tested and across different modalities, class-imbalance masking appears to be occurring, where models perform well in one class (in this case, Ideation) but perform only moderately in labelling the others. In particular, the majority of the wrong predictions under-classified the severity of the situation. However, we consider these results still promising to investigate a Screener Agent framework deployed in users' conversations with LLMs in a variety of settings, from personal usage to hospitals' virtual agents, where the LLM should be assessing the user's mental health for each message received. It is clear from our work that it will be essential to develop the correct risk levels and related definitions to ensure satisfactory performance.

There are important limitations affecting the work presented in this paper that should be acknowledged. Budgetary and time constraints restricted our team from performing an evaluation of some models (claude-sonnet-4.6, gemini-3.1-pro-preview) across all runs and from testing the modes described in 6.2 with the full Dataset 1. Another important limitation originating from Dataset 1 is the lack of a "neutral class", or a zero-risk class; the agents were not given the option to make the prediction that no suicide risk or topic was discussed. A future round of evaluations should examine how a Screener agent, looking out for suicidal ideation, would behave when presented with user messages that contain no suicide discussion. The lack of a zero-risk class, as well as the absence of domain experts on the research team, makes the current project unable to offer a strong uncertainty handling or escalation logic that could be used to create an Intervention Agent rather than a simple Screener. This will be an important step to explore in future work.

Another important limitation that resides in the dataset chosen (rsd_15k) is the dependency on human interpretation of its classes' descriptions. Many user messages, in fact, might sensibly belong to one class or the other, and medical professionals are well aware of the subjectivity of certain conclusions. While no perfect solution to this problem exist, it renders the task of suicide detection harder for LLMs, and it might explain why performance balanced across almost all models when the class probabilities mode was tested. Other minor challenges faced by the team were the lack of a licensed psychiatrist available to provide feedback for prompt adjustment to improve the results of the various evaluations, as well as the lack of expert labelling for the novel synthetic dataset present in this paper. Similarly, no fairness analysis was performed on Dataset 1 and 2 given the lack of expertise in the field; this is an aspect that should be explored in future work.

8.1 Future Work

There are multiple components of this project that could be further investigated in future work. With regards to the EvalRing framework, there is a lot of exciting potential for future improvements. Implementing an easy-to-use front-end dashboard for researchers to manage their models and experiment runs would provide non-AI experts a streamlined and approachable experience when researching AI in their field of study. EvalRing will be made available open-source in the near future by the team. We intend to keep exploring the efficacy of Screener Agents in a simulated 'Live Mode' setting, where a Patient Agent and a Chat Agent will simulate a conversation containing different levels of suicide intention, and the Screeners will be tasked to classify the risk in each of the user's messages, potentially with dedicated fine-tuning on rsd_15k. Verifying the validity of this Screener Agent framework is essential to contribute to a future practical application that might include different levels of threat escalation. Finally, the novel dataset (Dataset 2) allows for an exploration of the efficacy of utilizing synthetic datasets to replicate user messages in a medical study setting. Future work will be required to annotate by hand this dataset, as to better confirm the verisimilitude of the AI-generated messages.

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Appendix A Code & Contributions

A.1 Code & Data

The repository with the code that makes up the EvalRing framework, plus its application layer suicide_detection, can be found here: EvalRing Repository. The full results from all runs made for this paper are currently available at the following Google Drive folder:EvalRing Suicide Detection Results under "Results". The original dataset is publicly available via request to the original authors [19]. The cleaned-up dataset generated from the original one, as well as the novel dataset generated, cnn be found here: EvalRing Suicide Detection Results under "Datasets".

A.2 Contributions

The following is a breakdown of the contributions to the project from each author:

- **Luca Baiocchi**: Responsible for majority of related work research and writing the final report and other project components. Collaborated in data exploration, results analysis, and more.
- **Guan Huang**: Responsible for the EvalRing framework. Collaborated in related work research, data exploration, results analysis, and more.

Appendix B LLM Models Evaluated

B.1 Single Predictor Mode, Run 1

- command-a (Cohere) - $2.50/1M tokens (input), $10.00/1M tokens (output)
- gemini-3-flash-preview (Google) - $0.50/1M tokens (input), $3.00/1M tokens (output)
- glm-5 (Z-AI) - $1.00/1M tokens (input), $3.20/1M tokens (output)
- gpt-3.5-turbo (OpenAI) - $0.50/1M tokens (input), $1.50/1M tokens (output)
- gpt-4o (OpenAI) - $2.50/1M tokens (input), $10.00/1M tokens (output)
- grok-4.1-fast (X-AI) - $0.20/1M tokens (input), $0.50/1M tokens (output)
- kimi-k2.5 (Moonshot AI) - $0.60/1M tokens (input), $3.00/1M tokens (output)
- llama-3.3-70b-instruct (Meta) - $0.29/1M tokens (input), $2.25/1M tokens (output) via Cloudfare (Varies by provider)
- minimax-m2.5 (Minimax) - $0.30/1M tokens (input), $1.20/1M tokens (output)
- qwen3.5-plus-02-15 (Qwen AI)

B.2 Single Predictor Mode, Run 2

- deepseek-v3.2 - $0.28/1M tokens (input), $0.42/1M tokens (output)
- gemini-3-flash-preview (Google) - $0.50/1M tokens (input), $3.00/1M tokens (output)

- glm-5 (Z-AI) - $1.00/1M tokens (input), $3.20/1M tokens (output)
- gpt-3.5-turbo (OpenAI) - $0.50/1M tokens (input), $1.50/1M tokens (output)
- gpt-4o (OpenAI) - $2.50/1M tokens (input), $10.00/1M tokens (output)
- gpt-5-mini (OpenAI) - $0.25/1M tokens (input), $2.00/1M tokens (output)
- grok-4.1-fast (X-AI) - $0.20/1M tokens (input), $0.50/1M tokens (output)
- kimi-k2.5 (Moonshot AI) - $0.60/1M tokens (input), $3.00/1M tokens (output)
- llama-3.3-70b-instruct (Meta) - $0.29/1M tokens (input), $2.25/1M tokens (output) via Cloudfare (Varies by provider)
- minimax-m2.5 (Minimax) - $0.30/1M tokens (input), $1.20/1M tokens (output)
- qwen3.5-plus-02-15 (Qwen AI) - $0.40/1M tokens (input), $2.40/1M tokens (output)

B.3 Other Classification Modes (6.2)

- deepseek-v3.2 - $0.28/1M tokens (input), $0.42/1M tokens (output)
- gemini-3-flash-preview (Google) - $0.50/1M tokens (input), $3.00/1M tokens (output)
- gemini-3-pro-preview (Google) - $2.00/1M tokens (input), $12.00/1M tokens (output) (less than 200k tokens)
- gpt-3.5-turbo (OpenAI) - $0.50/1M tokens (input), $1.50/1M tokens (output)
- gpt-4o (OpenAI) - $2.50/1M tokens (input), $10.00/1M tokens (output)
- gpt-5-mini (OpenAI) - $0.25/1M tokens (input), $2.00/1M tokens (output)
- grok-4.1-fast (X-AI) - $0.20/1M tokens (input), $0.50/1M tokens (output)
- llama-3.3-70b-instruct (Meta) - $0.29/1M tokens (input), $2.25/1M tokens (output) via Cloudfare (Varies by provider)
- minimax-m2.5 (Minimax) - $0.30/1M tokens (input), $1.20/1M tokens (output)
- qwen3.5-plus-02-15 (Qwen AI) - $0.40/1M tokens (input), $2.40/1M tokens (output)

All costs are accurate as per the 14th of April, 2026.

Appendix C Additional Results

C.1 Additional Evaluation Graphs

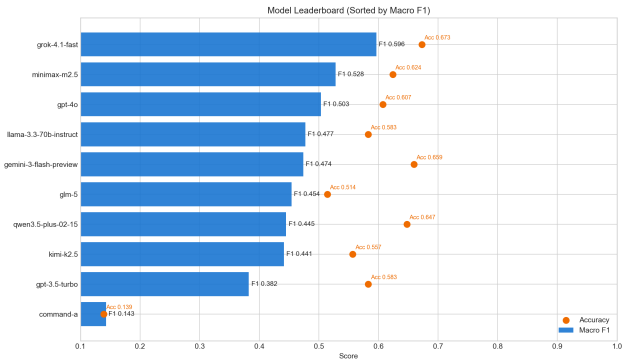
C.1.1 Single Predictor, Run 1

Model	Accuracy	F1 Score	Precision	Recall
*command-a**	0.139	0.143	0.377	0.090
gemini-3-flash-preview	0.660	0.474	0.490	0.485
glm-5	0.514	0.454	0.529	0.467
gpt-3.5-turbo	0.583	0.382	0.618	0.369
gpt-4o	0.607	0.503	0.588	0.499
grok-4.1-fast	0.673	0.596	0.614	0.608
kimi-k2.5	0.557	0.441	0.545	0.442
llama-3.3-70b-instruct	0.583	0.477	0.570	0.490
minimax-m2.5	0.624	0.528	0.543	0.536
qwen3.5-plus-02-15	0.648	0.445	0.485	0.448

Figure 5: LLMs as Suicide Screeners Over Four Risk Classes - Results Summary, Run 1. Best and Second Best in Green, Worst Performer in Red.

Model	Ideation	Indicator	Behaviour	Attempt
command-a*	0.203	0.3011	0.061	0.149
gemini-3-flash-preview	0.738	0.616	0.441	0.575
glm-5	0.571	0.220	0.220	0.503
gpt-3.5-turbo	0.717	0.345	0.239	0.229
gpt-4o	0.722	0.413	0.341	0.537
grok-4.1-fast	0.683	0.667	0.388	0.583
kimi-k2.5	0.662	0.534	0.139	0.430
llama-3.3-70b-instruct	0.709	0.289	0.348	0.563
minimax-m2.5	0.716	0.590	0.275	0.530
qwen3.5-plus-02-15	0.738	0.583	0.333	0.572

Figure 6: Single Prediction Mode - F1 Scores Per-Class Breakdown, Run 1.



(a) F1 Score Leaderboard, Run 1

	Average F1 Score (Run 1)*
Ideation	0.695
Indicator	0.473
Behaviour	0.303
Attempt	0.502

(b) F1 Score Per-Class Averages, Run 1. **command-a excluded.*

Figure 7: LLMs as Suicide Screeners Over Four Risk Classes - Single Prediction, Run 1

C.1.2 Single Predictor, Run 2

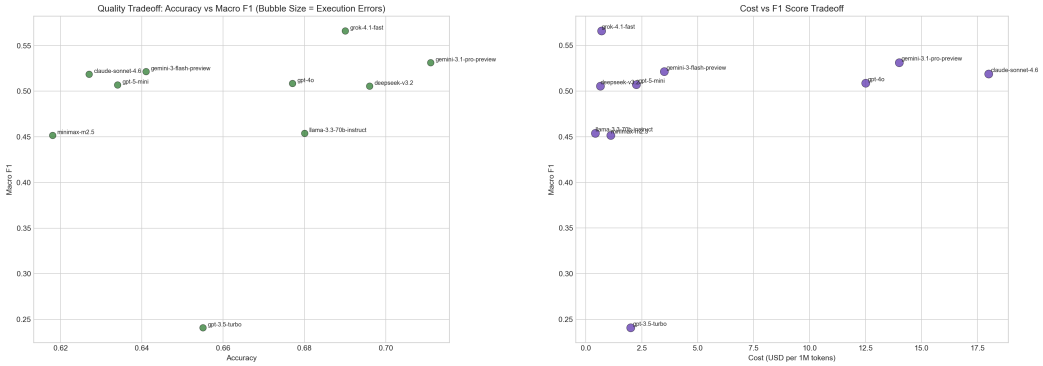
Model	Accuracy	F1 Score	Precision	Recall
deepseek-v3.2	0.593	0.500	0.566	0.534
gemini-3-flash-preview	0.657	0.478	0.501	0.485
glm-5	0.541	0.477	0.533	0.483
gpt-3.5-turbo	0.575	0.385	0.629	0.373
gpt-4o	0.603	0.508	0.603	0.503
gpt-5-mini	0.664	0.604	0.621	0.615
grok-4.1-fast	0.662	0.592	0.617	0.599
kimi-k2.5	0.550	0.349	0.441	0.349
llama-3.3-70b-instruct	0.575	0.473	0.559	0.481
minimax-m2.5	0.621	0.533	0.556	0.533
qwen3.5-plus-02-15	0.638	0.446	0.494	0.445

Figure 8: LLMs as Suicide Screeners Over Four Risk Classes - Results Summary, Run 2. Best and Second Best in Green, Worst Performer in Red.

Model	Ideation	Indicator	Behaviour	Attempt
deepseek-v3.2	0.709	0.376	0.468	0.447
glm-5	0.610	0.542	0.265	0.489
gemini-3-flash-preview	0.737	0.608	0.448	0.597
gpt-3.5-turbo	0.708	0.348	0.241	0.244
gpt-4o	0.717	0.401	0.350	0.564
gpt-5-mini	0.741	0.610	0.486	0.578
grok-4.1-fast	0.737	0.659	0.387	0.584
kimi-k2.5	0.655	0.533	0.147	0.408
llama-3.3-70b-instruct	0.705	0.289	0.338	0.560
minimax-m2.5	0.711	0.589	0.289	0.543
qwen3.5-plus-02-15	0.727	0.579	0.344	0.580

Figure 9: Single Prediction Mode - F1 Scores Per-Class Breakdown, Run 2.

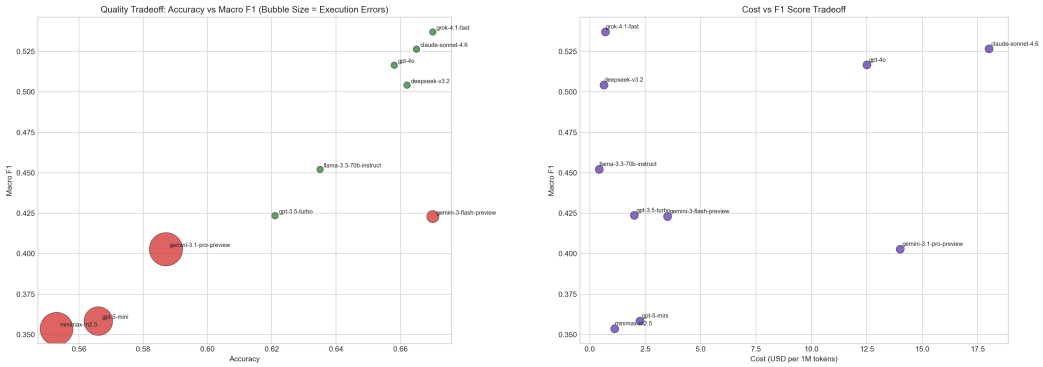
C.1.3 Multi-Class Probabilities



(a) F1/Accuracy Quality Tradeoff (b) F1/Cost Scatter Plot

Figure 10: Multi Class Evaluation - Additional Results Visualizations

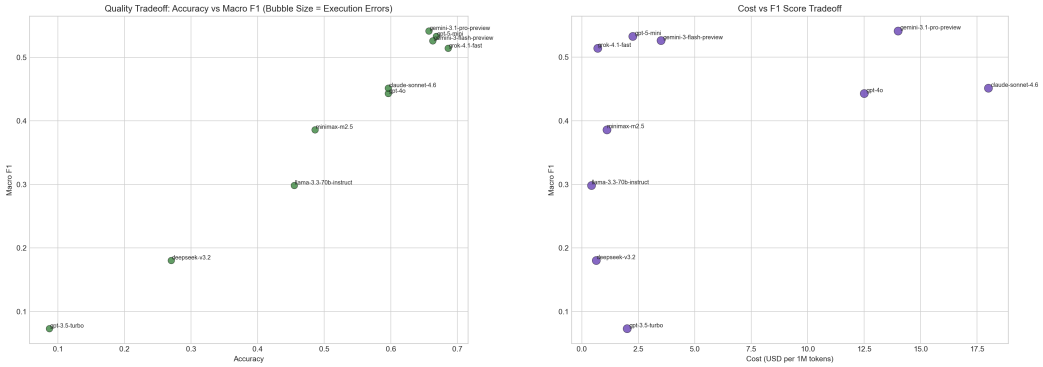
C.1.4 Rubric



(a) F1/Accuracy Quality Tradeoff (b) F1/Cost Scatter Plot

Figure 11: Rubric Evaluation - Additional Results Visualizations

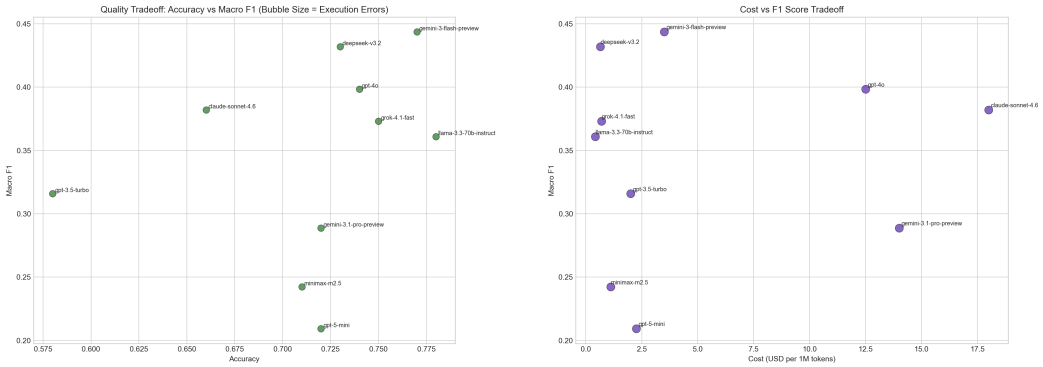
C.1.5 Comparative



(a) F1/Accuracy Quality Tradeoff (b) F1/Cost Scatter Plot

Figure 12: Comparative (Tournament) Evaluation - Additional Results Visualizations

C.1.6 LLM-as-Judge



(a) F1/Accuracy Quality Tradeoff (b) F1 Score Vs Cost Per Token

Figure 13: LLM-as-Judge Evaluation - Additional Results Visualizations

C.1.7 Novel Synthetic Dataset Evaluation Results

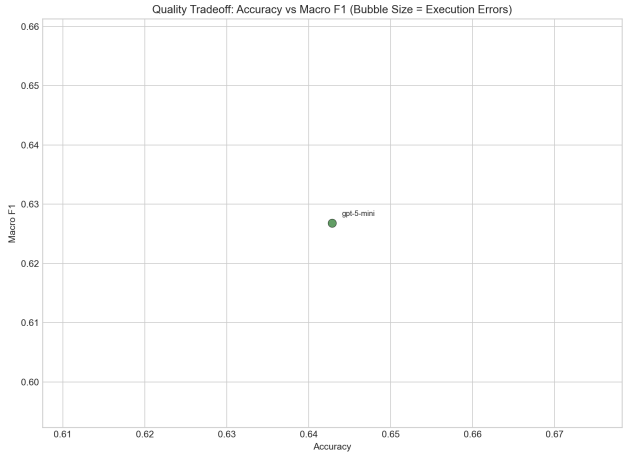


Figure 14: Synthetic Dataset Evaluation Results - F1/Accuracy Quality Tradeoff

C.2 Average Confusion Matrices

C.2.1 Single Predictor, Run 1 and Run 2

Actual \ Predicted	Ideation	Indicator	Behaviour	Attempt	Error
Ideation	4048.78	712.89	243.89	158.22	0.22
Indicator	757.56	1392.56	127.56	39.56	0.33
Behaviour	757.56	231.22	343.22	55.89	0.11
Attempt	88.33	67.00	81.56	268.00	0.11
Error	0.00	0.00	0.00	0.00	0.00

(a) Run 1 (*cohere-a excluded)

Actual \ Predicted	Ideation	Indicator	Behaviour	Attempt	Error
Ideation	4031.18	745.64	289.18	155.09	3.91
Indicator	1455.36	1411.27	147.09	46.82	2.36
Behaviour	763.55	193.73	431.09	66.64	1.18
Attempt	88.18	60.27	91.64	279.45	0.36
Error	0.00	0.00	0.00	0.00	0.00

(b) Run 2

Figure 15: Single Prediction Mode - Average Confusion Matrices

C.2.2 Multi-Class Probabilities & Rubric

Actual \ Predicted	Ideation	Indicator	Behaviour	Attempt	Error
Ideation	509.20	53.40	50.90	45.50	0.00
Indicator	106.10	103.10	4.90	0.90	0.00
Behaviour	34.10	3.80	16.60	9.50	0.00
Attempt	15.90	1.90	10.20	34.00	0.00
Error	0.00	0.00	0.00	0.00	0.00

(a) Multi Class Evaluation- Average Confusion Matrix

Actual \ Predicted	Ideation	Indicator	Behaviour	Attempt	Error
Ideation	451.60	57.50	60.50	77.90	11.50
Indicator	77.00	119.40	6.40	4.10	8.10
Behaviour	28.00	1.90	15.70	16.20	2.20
Attempt	8.20	0.10	10.80	42.00	0.90
Error	0.00	0.00	0.00	0.00	0.00

(b) Rubric Evaluation - Average Confusion Matrix

Figure 16: Multi-Class and Rubric Evaluations - Average Confusion Matrices

C.2.3 Comparative & LLM-as-Judge

Actual \ Predicted	Ideation	Indicator	Behaviour	Attempt	Error
Ideation	354.60	72.10	69.70	162.60	0.00
Indicator	74.30	105.20	18.50	17.00	0.00
Behaviour	19.90	3.40	12.30	28.40	0.00
Attempt	7.50	1.40	8.80	44.30	0.00
Error	0.00	0.00	0.00	0.00	0.00

(a) Comparative (Tournament) Evaluation- Average Confusion Matrix)

Actual \ Predicted	Ideation	Indicator	Behaviour	Attempt	Error
Ideation	61.50	3.10	4.80	2.60	0.00
Indicator	14.50	9.50	0.50	0.60	0.00
Behaviour	1.10	0.00	0.40	0.40	0.00
Attempt	0.30	0.10	0.60	0.10	0.00
Error	0.00	0.00	0.00	0.00	0.00

(b) LLM-as-Judge Evaluation- Average Confusion Matrix)

Figure 17: Comparative and LLM-as-Judge Evaluations - Average Confusion Matrices

C.2.4 Novel Synthetic Dataset Evaluation

Actual \ Predicted	Ideation	Indicator	Behaviour	Attempt	Error
Ideation	110	0	11	2	0
Indicator	80	69	7	6	0
Behaviour	17	27	100	46	0
Attempt	14	5	35	171	0
Error	0.00	0.00	0.00	0.00	0

Figure 18: Synthetic Evaluation Run - Confusion Matrix for gpt-5-mini

Appendix D Rubric for Rubric Classification Mode

The following Rubric was used to assign a score to each of the four suicide risk classes used in the Rubric Evaluation run.

- Score 0: "Wrong". The prediction is completely wrong — different category with no reasonable justification.
- Score 3: "Adjacent". "The prediction is in a neighboring/related category (e.g., Ideation vs Indicator). Understandable confusion but still incorrect.
- Score 5: "Partially Correct". The prediction captures some aspects of the correct category but misses key distinguishing features.
- Score 8: "Near Perfect". The prediction is very close — correct overall category, possibly a borderline case where both labels could be argued.
- Score 10: "Perfect". The prediction exactly matches the ground truth label.